⊆
a ≠ b
A ∩ B
A ∪ B
∪ A_n
∩ A_n
∏
∤

∅
∈
⊆
χ
τ
ε
η
$\alpha\beta\gamma\delta\varepsilon\zeta\eta\theta\iota\kappa\lambda\mu\varphi\phi$

==========
proof

QED
==========

Numbers

real numbers	abelian	R
integers	abelian	Z
Rational numbers	abelian	Q

Modular numbers

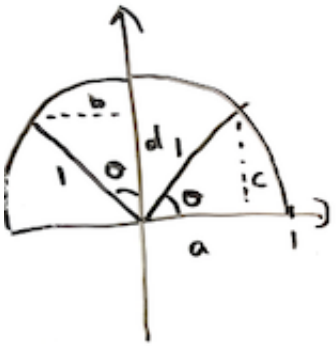
additive group mod n	abelian	Zn
multiplicative group mod p prime p	abelian	Zp* a^p-1 = 1 mod p Fermat's little theorem

Matrix

Let R be any commutative ring

General linear group over R	non-abelian	GL(n, R) = invertible n x n matrix i.e. det M ≠ 0
Special linear group with det M = 2	non-abelian	SL(n, R) = subgroup of GL(n, R)
Orthogonal group matrices	non-abelian	O(n, R) = n x n orthogonal
Special orthogonal group with det M = 2	non-abelian	SO(n, R) = subgroup of O(n, R)
aka rotation group	(SO(2) abelian)	e.g. SO(2) = { e^iθ = cos θ + iθ | any θ } = matrices of the form | cos θ -sin θ |

| sin θ cos θ |



$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

==========

== Small finite groups ==

n = 1	Z1	abelian		
n = 2	Z2	abelian		
n = 3	Z3	abelian		
n = 4	Z4	abelian		
	Z2 x Z2 ≃ D2	abelian	elements order	1, a, b, ab |a| = |b| = 2
n = 5	Z5	abelian		
n = 6	Z6 ≃ Z2 x Z3	abelian		
	Z3 ⋊ Z2		elements	1, a, b, ab, a^2b
	≃ D3		order	|a| = 3 |a^kb| = 2 for all k
			relation	**(a^kb)(a^kb) = 1** i.e. order 2 => a^kb = ba^-k
n = 7	Z7	abelian		
n = 8	Z8 = Z2 x Z2 x Z2	abelian		
	Z4 x Z2	abelian	elements order	1, a, b, ab |a| = |ab| = 4 |b| = 2
	Z4 ⋊ Z2 ≃ D4		elements order	1, a, b, ab, a^2b, a^3b |a| = 4 |a^kb| = 2 for all k

relation **(a^k·b)(a^k·b) = 1** i.e. order 2
=> a^k b = b a^-k

Z4 . Z2 elements 1, i, j
≃ Q4 order |i| = |j| = 4

relation **i^2 = j^2**
ji = i^-1 j
=> |ij| = 4

Or the traditional -1, i, j, k where -1, k are derived from i, j by

-1 := i^2 = j^2 **k := ij**

=> i^2 = j^2 = k^2 = ijk = -1

n = 9	Z9	abelian	
	Z3 x Z3	abelian	1, a, b, ab |a| = |b| = 3
n = 10			

Quaternion group
The quaternion group Q8 is defined by -1, i, j, k where

order
|-1| = 2
|i| = |j| = |k| = 4

relation
-1 = i^2 = j^2 = k^2
= ijk

Equivalently, this implies that -1, k can be defined in terms of i, j

-1 := i^2 = j^2

k := -(j^-1)(i^-1) = (j^2)(j^3)(i^3)
= ji^3

and

==========

== Symmetric group ==

Symmetric group
For any set X, the **symmetric group S(X)** is the set

S(X) := set of bijections X → X
aka permutation

Sn := S(Zn)

Permutations σ, τ are disjoint if their fixed points are disjoint

fixed point x $\sigma(x) = x$

disjoint σ, τ x is a fixed point in at least one of σ or τ
for any $x \in X$

A **cycle** is a permutation which is fixed everywhere except for a single sequence.
i.e. σ is fixed except on a single sequence

i, σ(i), σ^2(i), σ^3(i), ...

σ(j) = σ^l(i) for some l
j otherwise

The symmetric group is a type of **free** structure.
i.e. if G is a group, then

g *: G → G $\in$ S(G)
g * x = gx

i.e. Im [*: G → G] is a subgroup of S(G)

is an **embedding of G** in S(G).

Cycle representation
A **cyclic permutation** can be represented by a sequence

(i σ(i) σ^2(i) σ^3(i) ....)

A **transposition** is a cycle (a b) of order 2 i.e. flips a and b

σ = (a b) <=> σ(a) = b σ(b) = a
σ(x) = x otherwise

Note: **composition** of permutation is read from **right to left** since they are function composition
e.g. if σ = (a b) τ = (a c)

$\tau \circ \sigma$(i) = $\tau(\sigma$(i))

(a c) (a b) = (a b c)
b <- b b <- a b <- a
c <- a a <- b c <- b
a <- c c <- c a <- c

Cycles are important because they
i.e. given any σ

σ = $\prod$ (ai1 ai2 ...) unique product of **disjoint cycles**

= $\prod$ (ai1 ai2)(ai2 ai3)(ai3 ai4) ... product of **transpositions**

Matrix representation

Permutation can be represented by a matrix

$\sigma \in Sn$ $\quad m\sigma = | e\sigma(1)\ e\sigma(2)\ e\sigma(3)...|$ $\quad$ i.e. **column** represents **input i**

row = result $\sigma(i)$

$= \{xij\}$

where **xij = 1** if $i = \sigma(j)$

Composition

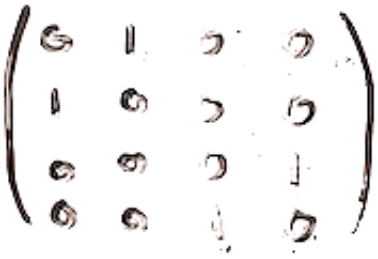
$\pi\sigma$ $\quad$ <=> $\quad$ **m**$\pi \cdot$ **m**σ

e.g. permutations of S4

(1 2 3 4)



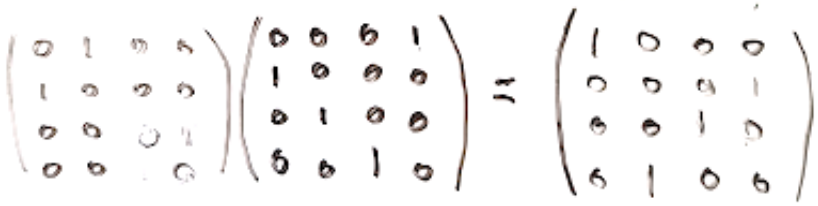
(1 2)(3 4)



(1 2)



(1 2)(3 4) · (1 2 3 4) = (2 4)



Basic properties

Theorem: basic properties

Basic properties

commutative

σ, τ **disjoint** $\quad$ => $\quad \sigma\ \tau$ **=** $\tau\ \sigma$

But clearly disjoint is NOT necessary.

σ, τ disjoint <=> $\sigma(\tau(i)) = \tau(\sigma(i))$ for all i

<=> for any j, k

$\sigma(j) = \tau(k)$ <=> τ^-1(j) = σ^-1(k)

e.g.

σ	$\tau = \sigma$^-1	inverse
σ = (1 2)(3 4)	τ = (1 2)(5 6)	common transpose (1 2)
σ = (1 2)(3 4)	τ = (1 3)(2 4)	non-trivial

Hence, the only fully commutative element is the identity

Z(Sn) = { 1 }

fixed point

i is a fixed point of σ => τ^-1(i) is a fixed point of σ^τ

Let

Fix(i) = { σ(i) = i: $\sigma \in$ S }
i.e. set of permutations fixing i

Then

Fix(i) is a subgroup

Fix(j) = Fix(i)^τ for any τ(i) = j
i.e. Fix(i) are all **conjugates**

Simply by counting

Fix(i)^τ $\subseteq$ Fix(j)
Fix(j)^τ^-1 $\subseteq$ Fix(i)

=> |Fix(i)| = |Fix(i)^τ| <= |Fix(j)| = |Fix(j)^τ^-1| <= |Fix(i)|

=> |Fix(i)| = |Fix(j)|

transpose

(a b)	= (b a)	
(a b)(a b)	= 1	**inverse**
(x y)(a b)	= (a b)(x y)	**disjoint** product
(b x)(a x)	= (b x a)	**intersection** (before x after) = (before x)(after x)

Hence

(a b c)	= (a c)(b a)	intersect at a

= (a b)(b c) intersect at b
= (b c)(a c) intersect at c

Cycle

(i j) (1... i.... j....) = (1...i-1 j...) (i...j-1)
(1... i.... j....) (i j) = (1...i j+1...) (i+1 ...j)
(i j) (1... i.... j....) (i j) = (1...i-1 j i+1....j-1 i ...)

Conjugation

(a x)(b x)(a x) = (a x)(a x)(a b)
= (a b)
= (b x)(a x)(b x)

σ (a b) σ^-1 = (σ(a), σ(b)) for any σ

generator

Hence both are generators

(a 1), (a 2), (a 3) ... since for any x, y
for any a (x y) = (a x)(a y)(a x)
= (a y)(a x)(a y)

(1 2), (2 3), (3, 4) ... since for any x < y
(x+2 x) = (x+1 x+2)(x+1 x)(x+1 x+2)
(x+3 x) = (x+2 x+3)(x+2 x)(x+2 x+3)
...
(y x) = (y-1 y)(y-1 x)(y-1 y)

decomposition

cycle

A cycle is a product of transpositions

(a b c d e) = **(a b c d)(a e)**
= (a b c)(a d)(a e)
...
= **(a b)(a c)(a d)(a e)**

(a b c d e) = **(d e)(a b c d)**
= (d e)(c d)(a b c)
...
= (d e)(d c)(c b)(b a)

k cycle => product of **(k - 1)** transposes

unique decomposition

Any permutation σ is a product of **unique disjoint cycles**

σ = $\prod$ (ai1 ai2 ai3 ...)
unique upto cycle order

Hence a product of **transposes**, not necessarily unique

σ = $\prod$ (ai1 ai2)(ai2 ai3)(ai3 ai4) ...

= ∏ ... (ai1 ai4)(ai1 ai3)(ai1 ai2) another decomposition

i.e. not unique

Product of cycles

Let σ, τ be cycles.

If for some i, i'

X = σ(i) = τ(i')

then

σ τ σ

$\tau \circ \sigma$ = (... σ^-1(x) τ(x) τ^2(x) .... τ^-1(x) **X** σ(x) σ^2(x) ...)

$\sigma \circ \tau$ = (... σ^-1(x) **X** τ(x) τ^2(x) .... τ^-1(x) σ(x) σ^2(x) ...)

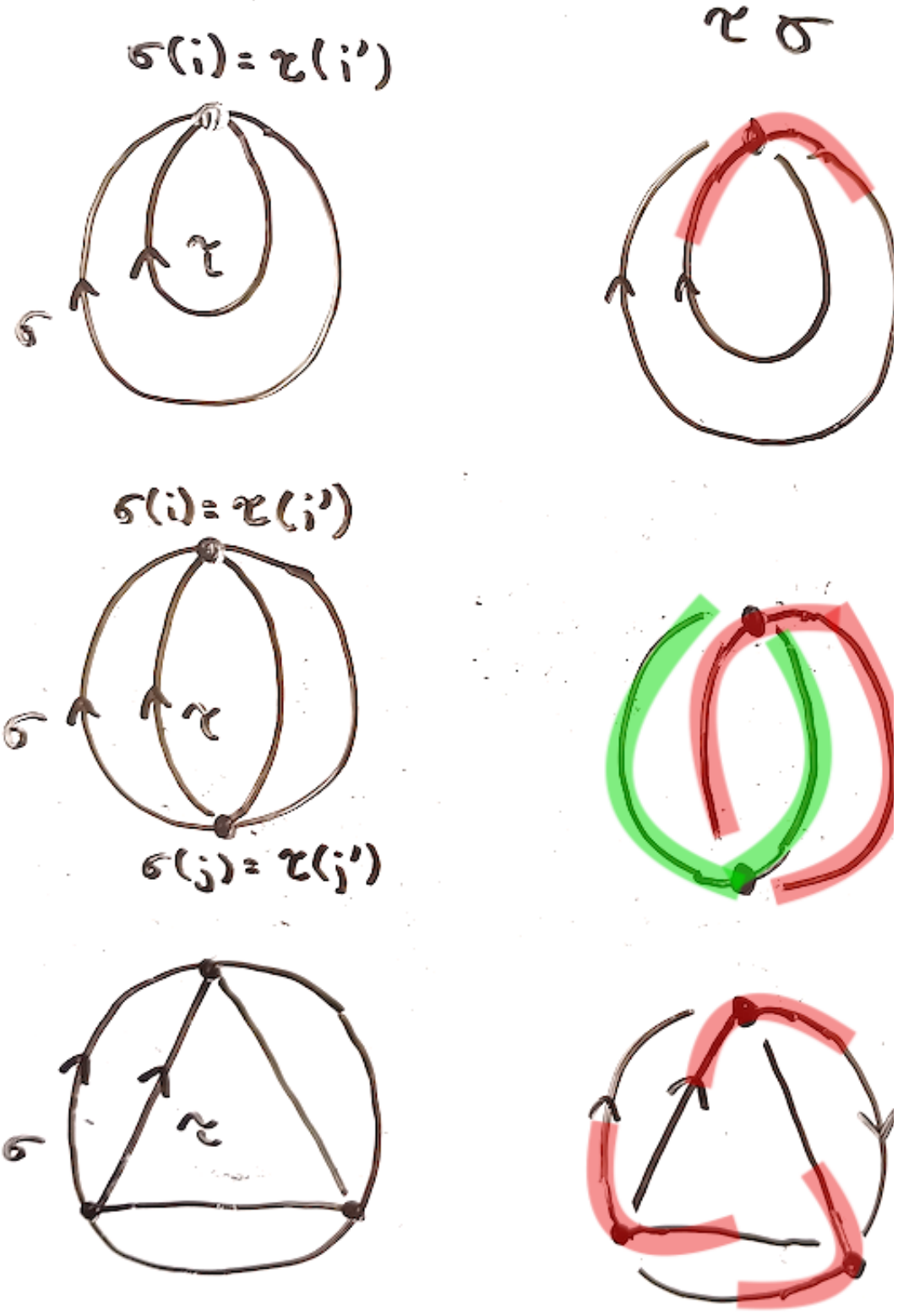
i.e. the product connects at the intersection.
Hence the product is determined by the **number of intersections**.

|$\sigma \cap \tau$| is **odd** => 1 cycle

even => 2 disjoint cycles

e.g.

# intersection	**cycles**	**product**
1	σ = (a b ★ c d) τ = (1 ★)	join $\tau \circ \sigma$ = (a b 1 ★ c d) $\sigma \circ \tau$ = (a b ★ 1 c d)
1	σ = (a b ★ c d) τ = (1 2 3 ★ 4)	join $\tau \circ \sigma$ = (a b 4 1 2 3 ★ c d) $\sigma \circ \tau$ = (a b ★ 4 1 2 3 c d)
2	σ = (a b ★ c d ▲ e f) τ = (1 ▲ 2 3 ★ 4 5)	split $\tau \circ \sigma$ = (a b 4 5 1 ▲ e f) (c d 2 3 ★) $\sigma \circ \tau$ = (a b ★ 4 5 1 e f) (c d ▲ 2 3)

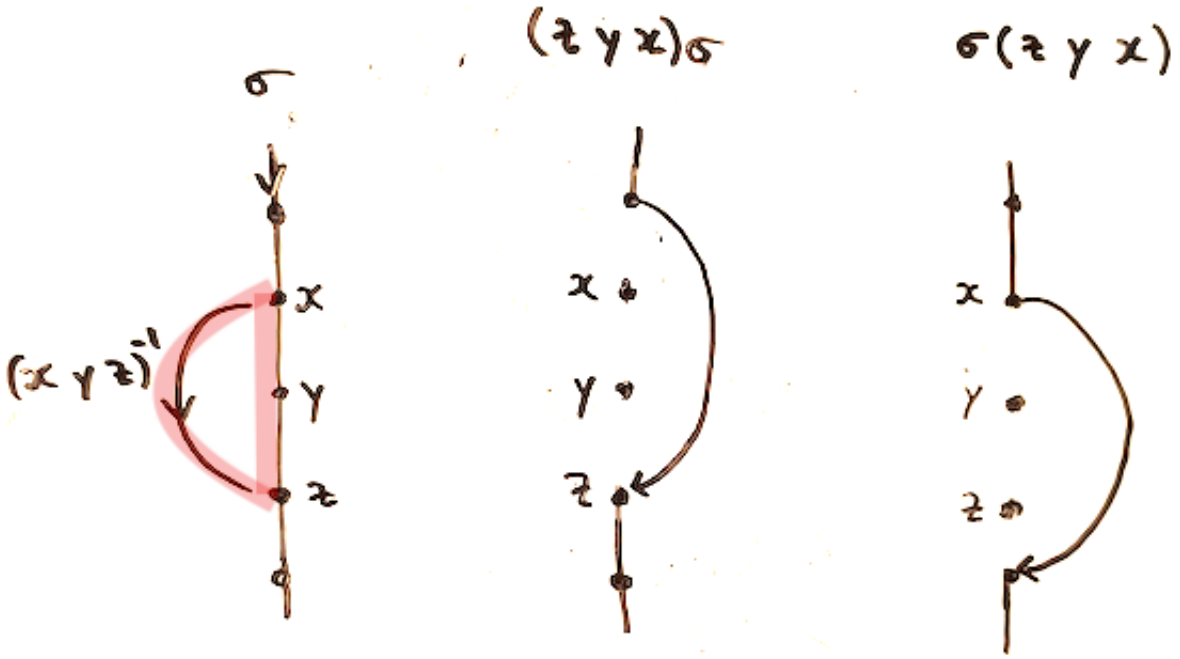


A special case is when one is **embedded** in the other.
e.g.

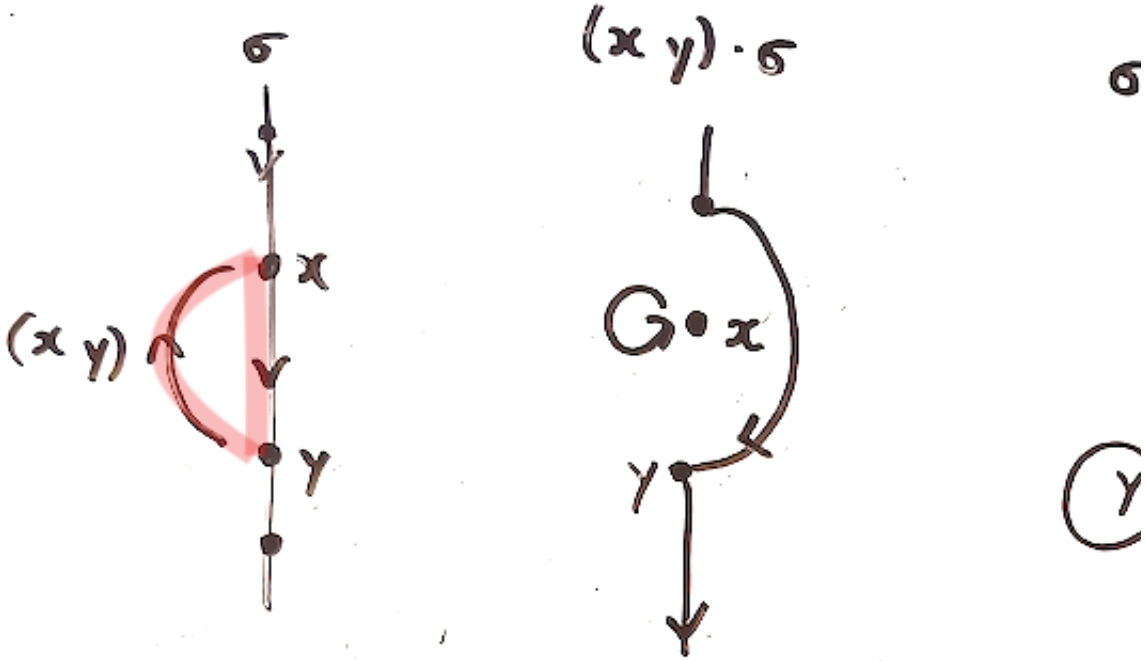
σ = (... a b c ....)
τ = (a b c)

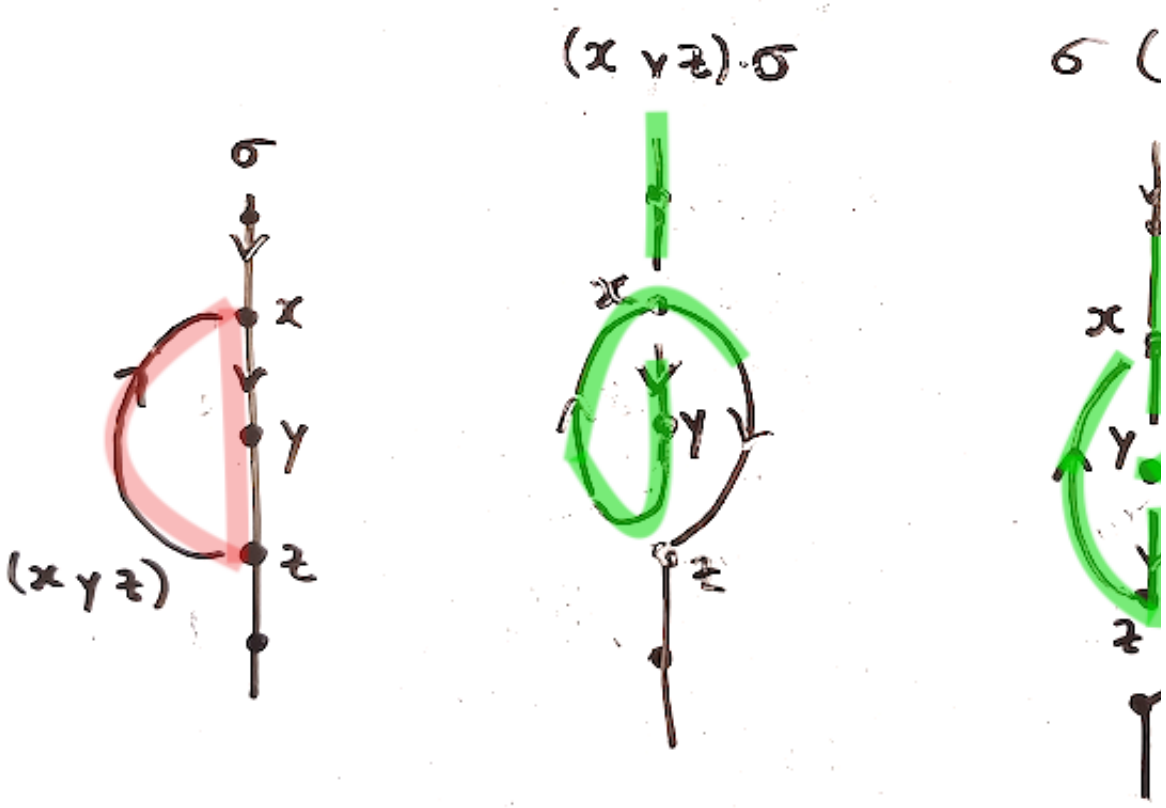
In particular, if the embedding is an **inverse**

(a1 a2 a3 ... an)^-1 σ = σ with **a1, a2, ... an-1** removed
σ **(a1 a2 a3 ... an)^-1** = σ with **a2, a3, ... an** removed



TODO





Derangements
https://en.wikipedia.org/wiki/Derangement
TODO

Alternating group

https://en.wikipedia.org/wiki/Alternating_group

Even/odd
Alternating group
Sign
https://en.wikipedia.org/wiki/Parity_of_a_permutation
σ **inverts i and j** if

i < j but π(i) < π(j)

The **sign** of a permutation is the parity of the number of inversions

sgn σ = -1 odd # inversions
1 even # inversions

= (-1^)^# inversions

The **sign** of a permutation represents the number of transposes in the decomposition.

		# of transposes
sgn(σ)	=	-1 **odd**
		1 **even**

= (-1)^ (# of transposes)

Hence

sgn($\sigma \circ \tau$) = sgn(σ) sgn(τ)

sgn(σ^-1) = sgn(σ)

Some specific values

sgn(1)	= 1	
sgn((x y))	= -1	transpose
sgn(k-**cycle**)	= (-1)^(k - 1)	cycle

The **alternating group An** of Sn is the group of even permutations

An := { $\sigma \in$ Sn | sgn(σ) = 1 }

normal subgroup
since sgn($\tau\sigma\tau$^-1) = sgn(τ)^2 sgn(σ) = sgn(σ)

Theorem: sign definitions
Equivalent formulations of sgn: Sn → { 1, -1 }

a) **inversions**

sgn σ = (-1)^# of inversions

Or an explicit formula:
The **vandermonde polynomial** is

p(x1, x2, ...) = $\prod_{i<j}$ (xi - xj)

Inversions are then tracked by

sgn σ = p(xσ(1), xσ(2), ...) / p(x1, x2, ...)

b) **transpose decomposition**

sgn σ = (-1)^# of transposes in decomposition

c) **constructive**

sgn(x y) = -1

sgn $\tau\sigma$ = sgn τ · sgn σ

==========

proof

case: transpose => constructive
trivial

case: inversion => constructive

sgn(x y) = -1

sgn($\sigma\tau$) = p(x_$\sigma\tau$(1), x_$\sigma\tau$(2), ...) / p(x_1, x_2, ...)
= p(x_$\sigma\tau$(1), x_$\sigma\tau$(2), ...) / p(x_τ(1), x_τ(2), ...)
* p(x_τ(1), x_τ(2), ...) / p(x_1, x_2, ...)
= sgn(σ) sgn(τ)

QED
==========

Theorem: sign properties

multiplicative **sgn** $\sigma\tau$ **= sgn** σ ∘ **sgn** τ

basic values

identity	sgn (1)	= 1
transpose	sgn (x,y)	= -1
k-cycle	sgn (k cycle)	= (-1)^(k+1)

Theorem: Alternating group
The alternating group is simple except in S4

==========

proof
Let N be a normal subgroup not { 1 }
Idea: show that N contains all **3 cycles** (a b c), and hence generates An

Step 1: An contains at least **3 cycles** (a b c) or **2 transpositions** (a b)(c d) from x.
Let x ∈ An and consider its cycle decomposition.
Then x should have at least one of the following forms

... (1 2 3 4 ... n) ...	n > 3
... (1 2 3) ...	3 cycles
... (1 2)(3 4) ...	2 transpositions

The idea is to **cancel** the outer dots by modifying the inner part with

case	**y = (3 2 1)x(1 2 3)**	**xy^-1**
n > 3	...(1 2 4 ... n 3) ...	(1 4 3)
3 cycle	...(3 2 1)...	(3 2 1)
2 tranpositions	...(1 3)(2 4) ...	(1 4)(2 3)

Step 2: Reduce (a b)(c d) to a 3 cycle
IMPORTANT: this needs at least **5 elements** i.e. at least **S5**

Let x = (1 2)(3 4)

y = (5 4 3)x(3 4 5)
= (1 2)(3 5)

=> xy = (3 4)(3 5)
= (3 4 5) ∈ An

Step 3: An contains all 3 cycles
Suppose x = (1 2 3) ∈ An

swap 2 -> a	(1 2 a)x(a 1 2) = (1 3 a)
change order	x = (1 2 3) x^2 = (3 2 1)

QED
==========

Regular

Regular
https://en.wikipedia.org/wiki/Group_action
A subset H ⊆ Sn is **regular** if H is **free** and **transitive**.
i.e. there is a unique

h(i) = j

for every i, j.

Equivalently, |H| = n and

$$h(i) \neq h'(i)$$

for all i and $h \neq h'$.
So if H is labeled by some fixed base point 0

$$i = hi(0)$$

then for any, i, j

$$\mathbf{hj \cdot hi^{-1}(i) = j}$$

i.e. H is determined by **h1,... , hn**

e.g. S4
claim: If $H \subseteq S4$ is regular, then H is determined by

h1 order 4
h2 = h1^2

Hence

H ≃ Q4 i.e. quarternion group

==========

proof
If $H \subseteq S4$ is regular, and label H as

i = hi(1) i = 1 ... 4

If

h1 h2^-1 = h1^-1 (1)
h3 h1^-1 = h2 (2)
h3 = h1^-1

then

h2 = h1^2 by 1
= h1^-2 by 2

=>

h2^2 = 1
h1^4 = 1

i.e. h1, h2 have order 4, 2 respectively.
Then

1 -> 2 h2· h1^-1 = h1
1 -> 3 h3· h1^-1 = h1^-2 = h2
2 -> 1 h1· h2^-1 = h1^-1
2 -> 3 h3· h2^-1 = h1
3 -> 1 h1· h3^-1 = h1^2 = h2
3 -> 2 h2· h3^-1 = h1^-3

with no contradiction

1, 2 (h2· h1^-1) · (h1· h2^-1) = h1 · h1^-1 = 1
1, 3 (h3· h1^-1) · (h1· h3^-1) = h2 · h2 = 1
2, 3 (h3· h2^-1) · (h2· h3^-1) = h1 · h1^-3 = 1

QED
==========

Sn

	elements	subgroup
S1	1	C1 {1}
S2	1	C1 {1}
order: 2! = 2	(1 2)	C2 {1, (1 2)}
S3	1	C1 {1}
order: 3! = 6	(1 2)	C2 x 3 {1, (1 2)} {1, (1 3) }, {1, (2 3)}
	(2 3)	C3 {1, (1 2 3), (1 3 2) }
	(1 3)	
	(1 2 3) = (1 3)(1 2)	
	(1 3 2) = (1 2 3)^-1	
S4	1	C1 {1}
order: 4! = 24	(1 2)	C2 x 6 {1, (1 2)}, {1, (1 3)}, ...
	(1 3)	C3 x 4 {1, (1 2 3), (1 3 2)}, {1, (1 2 4), (1 4 2)}, ...
	(1 4)	C4 x 3 {1, (1 2 3 4), (1 3) (2 4), (1 4 3 2)}, {1, (1 2 4 3), (1 4)(2 3), (1 3 4 2)} {1, (1 3 2 4), (1 2)(3 4), (1 4 2 3)}
	(2 3)	A4 {1, (1 2 3), (1 2 4), ...3-cycles..., (1 2)(3 4), (1 3)(2 4), (1 4)(2 3)}
	(2 4)	even subgroup {1, (1 2)(3 4), (1 3)(2 4), (1 4)(2 3)}
	(3 4)	
	(1 2 3) = (1 3)(1 2)	
	(1 2 4) = (1 4)(1 2)	
	(1 3 2) = (1 2)(1 3) = (1 2 3)^-1	
	(1 3 4) = (1 4)(1 3)	
	(1 4 2) = (1 2)(1 4) = (1 2 4)^-1	
	(1 4 3) = (1 3)(1 4) = (1 3 4)^-1	
	(2 3 4) = (2 4)(2 4)	
	(2 4 3) = (2 3)(2 5) = (2 3 4)^-1	
	(1 2 3 4) = (1 4)(1 3)(1 2)	
	(1 2 4 3) = (1 3)(1 4)(1 2)	
	(1 3 2 4) = (1 4)(1 2)(1 3)	
	(1 3 4 2) = (1 2)(1 4)(1 3) = (1 2 4 3)^-1	
	(1 4 2 3) = (1 3)(1 2)(1 4) = (1 3 2 4)^-1	
	(1 4 3 2) = (1 2)(1 3)(1 4) = (1 2 3 4)^-1	
	(1 2)(3 4)	
	(1 3)(2 4)	
	(1 4)(2 3)	

==========

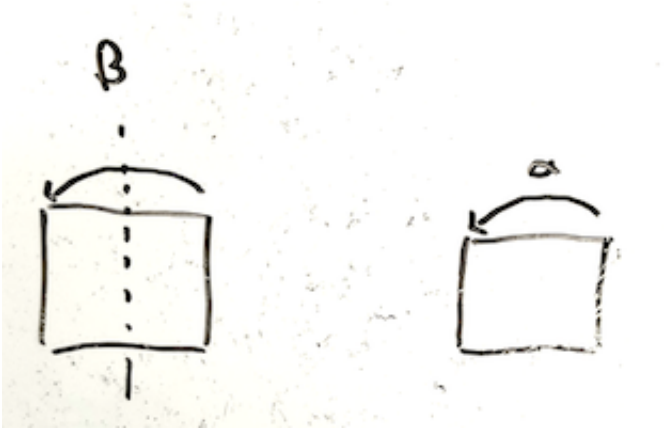
== Dihedral group ==

Dihedral group
https://en.wikipedia.org/wiki/Dihedral_group
Special case of symmetric group.
Symmetric operations on a **regular polygon** with **n** sides

Proposition
It is described completely by:

a) Dn is generated by two operations:

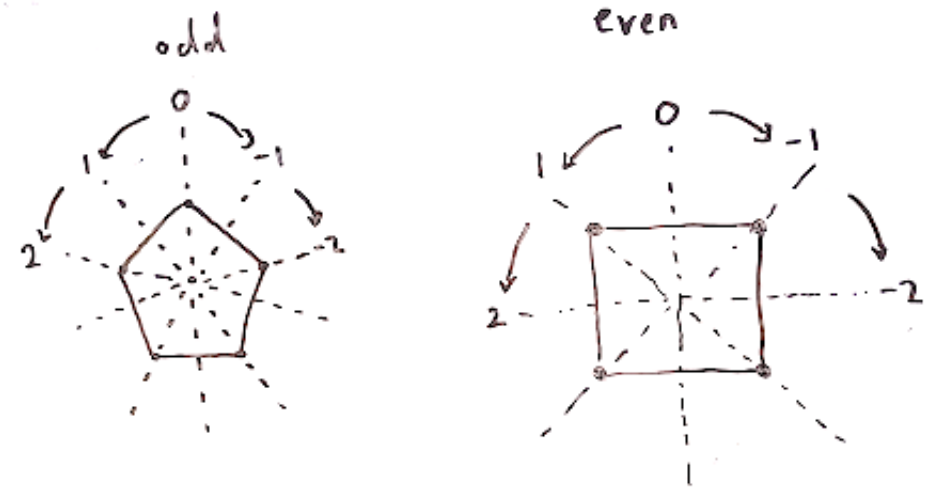
α = rotation
β = reflection along an axis



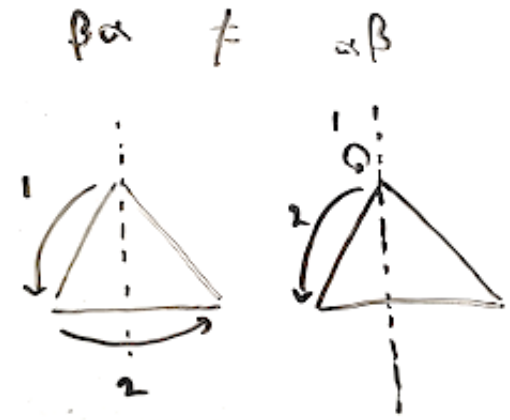
a) **|Dn| = 2n** since

n rotations n vertices

n reflections if n is odd, n reflection axes through **vertex** (one per vertex)
if n is even, n/2 reflection axes through **opposing vertices** (n/2 pairs of vertices)
n/2 reflection axes through **opposing sides** (n/2 pairs of sides)



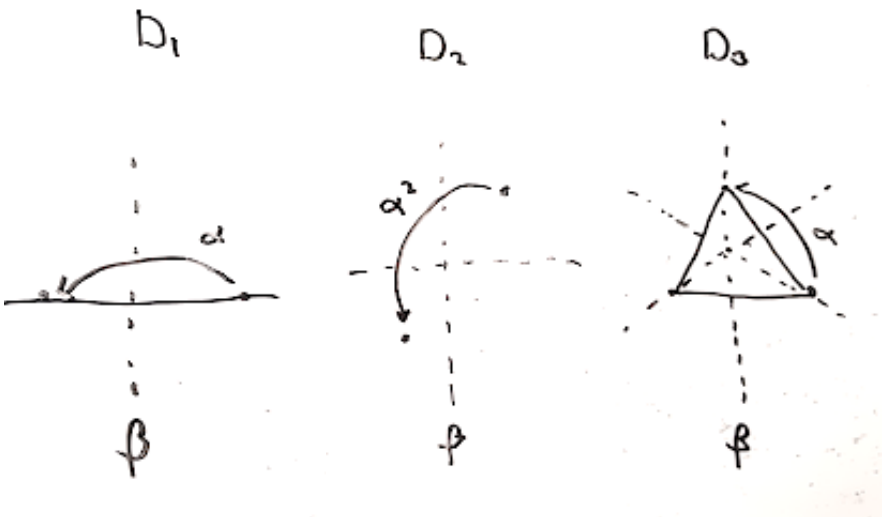
c) **non-abelian** e.g. if x is a fixed vertex of β i.e. $\beta(x) = x$ then
$\alpha\beta(x) = \alpha(x)$
$\beta\alpha(x) != \alpha(x)$

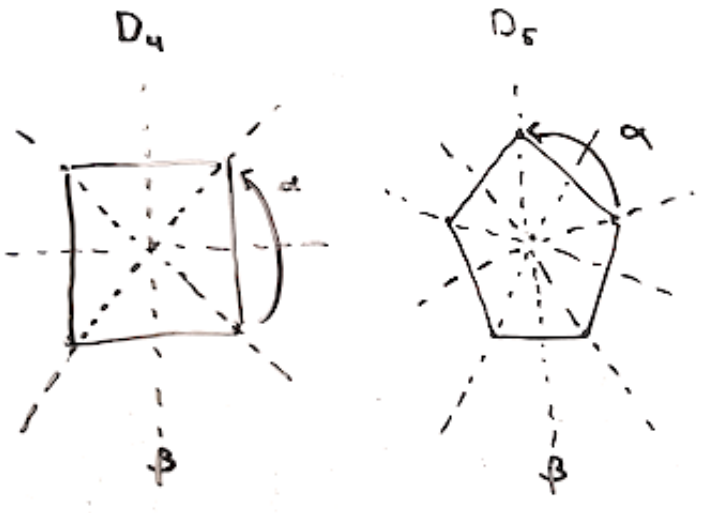


Examples

D1 = { 1, β } D2 = { 1, α, β, $\alpha\beta$ } **D3** = { 1, α, α^2, β, $\alpha\beta$, $\alpha^2\beta$ } **D4** = { 1, α, α^2, β, $\alpha\beta$, $\alpha^2\beta$, $\alpha^3\beta$ } **D5** = { 1, α, α^2, β, $\alpha\beta$, $\alpha^2\beta$, $\alpha^3\beta$, $\alpha^4\beta$ }

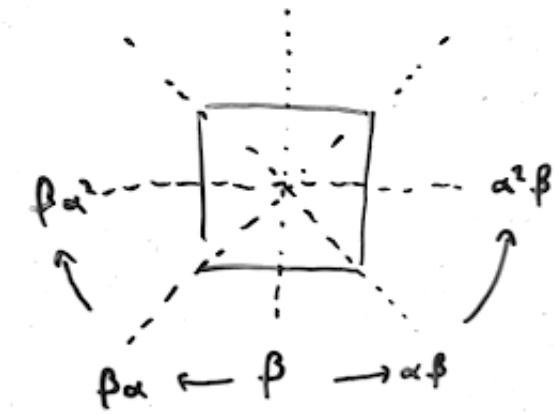
α	-	rotation by π/2	rotation by π/3
	rotation by π/4		rotation by π/5
$\alpha\beta$	-	β rotated by π/2	β rotated by π/6
	β rotated by π/8		β rotated by π/10
$\alpha^2\beta$	-	-	β rotated by π/3
	β rotated by π/4		β rotated by π/5
$\alpha^3\beta$	-	-	-
	β rotated by 2π/8		β rotated by 3π/10





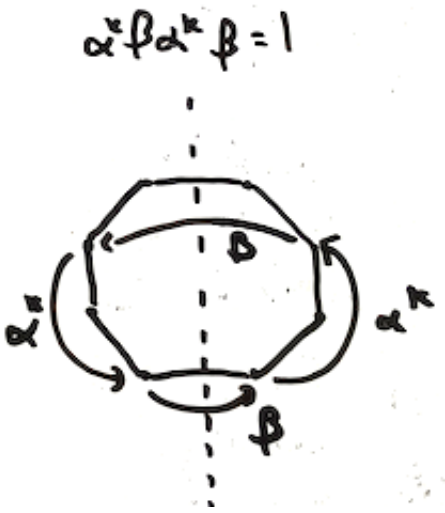
Theorem: basic identities

α^k		**k** rotations (anticlockwise)
α^n	= 1	
β		**reflection**
β^2	= 1	
$\alpha^k\beta$		**reflection** axis rotated **k times**
$\beta\alpha$^k	= α^-kβ	**reflection** axis rotated **-k times i.e. clockwise**



Hence the fundamental identity

$\alpha^k\beta$ $\alpha^k\beta$ = 1 **fundamental identity**



$\alpha^k\beta$	$= \beta\alpha^{-k}$	**commute**
$(\alpha^k\beta)\ (\alpha^l\beta)$	$= \alpha^k(\alpha^{-l}\beta)\beta$	**reflection** by k ◦ **reflection** by l = **rotation** by (k - l)
	$= \alpha^{(k-l)}$	
$\alpha^k\ (\alpha^l\beta)$	$= \alpha^{(k+l)}\beta$	**rotation** by k ◦ **reflection** by l = **reflection** by (k + l)
$(\alpha^l\beta)\ \alpha^k$	$= \alpha^{(k-l)}\beta$	**reflection** by l ◦ **rotation** by k = **reflection** by (k - l)

conjugates

	by rotation α^k α^k x α^{-k}	**by reflection** $\alpha^k\beta$ $(\alpha^k\beta)$ x $(\alpha^k\beta)$
rotation x = $\alpha^{}$,	rotation x $= \alpha^m$	rotation x^-1 $= \alpha^{-m}$
reflection x = $\alpha^m\beta$	reflection α^{2k} x $= \alpha^{m+2k}\ \beta$	reflection $\alpha^{2(k-m)}$ x $= \alpha^{2k-m}\ \beta$
x = β	reflection α^{2k} x	reflection α^{2k} x

conjugacy classes
The conjugacy classes are

1) 1^Dn = { 1 }

2) for each k

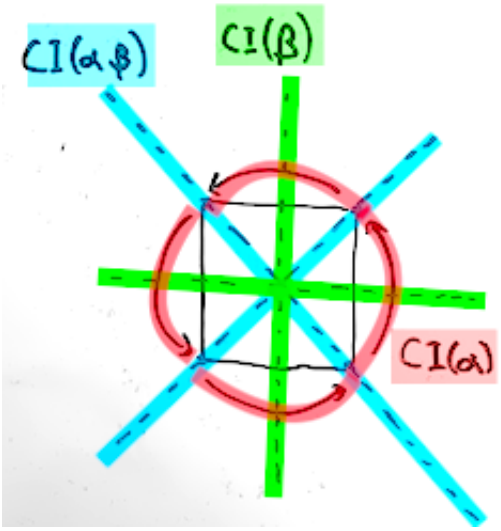
$(\alpha^k)^{Dn} = \{\alpha^k, \alpha^{-k}\}$ since
$\beta\ \alpha^k\ \beta = \alpha^{-k}$
$(\alpha^l\beta)\ \alpha^k\ (\alpha^l\beta)^{-1} = \alpha^{-k}$

3) $\beta^{Dn} = \{\ \alpha^{2k}\beta \mid \text{all k}\}$ since
$\alpha^k\beta\ \alpha^{-k} = \alpha^{2k}\beta$
$(\alpha^k\beta)\ \beta\ (\alpha^k\beta)^{-1} = \alpha^{2k}\beta$

4) $\alpha\beta$^Dn = { α^(2k+1)β | all k}
β^Dn = $\alpha\beta$^Dn if n is odd

Hence

n odd	1, α, ..., α^(n-1)/2, β	(n + 3) / 2 classes
n even	1, α, ..., α^n/2, β, $\alpha\beta$	(n + 6) / 2 classes



center

Z(Dn) = { 1 } n odd

(α^n/2) n even
= { 1, α^n/2} i.e. cyclic group of π/2 rotation

since
α^m α = α α^k
α^m β = β α^-m => commutes only if m = -m (mod n) i.e. m = 0, n/2

β α^1 = α^-1 β => reflection β never commutes with α

Representation
a) Abstract representation as **semi-product** Ck ⋊ C2

(k, 0)	**rotation** k times
(k, 1)	**reflection**

inverse

(k, 0)^-1	=	(k, 0)	rotation
(k, 1)^-1	=	(-k, 1)	reflection

product

(k, 0) (l, 0)	=	(k + l, 0)	rotation ◦ rotation = rotation
(k, 0) (l, 1)	=	(k + l, 1)	rotation ◦ reflection = reflection
(k, 1) (l, 0)	=	(k - l, 1)	reflection ◦ rotation = reflection
(k, 1) (l, 1)	=	(k - l, 0)	reflection ◦ reflection = rotation

b) **Group representation** GL(Zn)

i.e. permutations of Zn

e.g. D4
Basis

α = 0 1 0 0 **rotation**
0 0 1 0
0 0 0 1
1 0 0 0

β = 0 0 0 1 **reflection**
0 0 1 0
0 1 0 0
1 0 0 0

$\alpha\beta$ = 0 0 1 0
0 1 0 0
1 0 0 0
0 0 0 1

e.g. D5
Reflection has a fixed point

α = 0 1 0 0 0 **rotation**
0 0 1 0 0
0 0 0 1 0
0 0 0 0 1
1 0 0 0 0

β = 1 0 0 0 0 **reflection**
0 0 0 0 1 fixed point at i=0
0 0 0 1 0
0 0 1 0 0
0 1 0 0 0

$\alpha\beta$ = 0 0 0 0 1
0 0 0 1 0
0 0 1 0 0 fixed point at i=3
0 1 0 0 0
1 0 0 0 0

c) Geometric representation

α = cos 2π/n - sin 2π/n
sin 2π/n cos 2π/n

β = 1 0
0 -1

<u>Subgroups</u>

A dihedral group Dn has two types of subgroups
For each r = n/m for m | n

Rotation subgroup Zm
(α^r) ~ Zm **cyclic** subgroup generated by rotation

normal

Reflection subgroup Z2 = D1
(β) **self normal** if n odd (product subgroup with m = 1)
($\alpha\beta$)
.... r variants

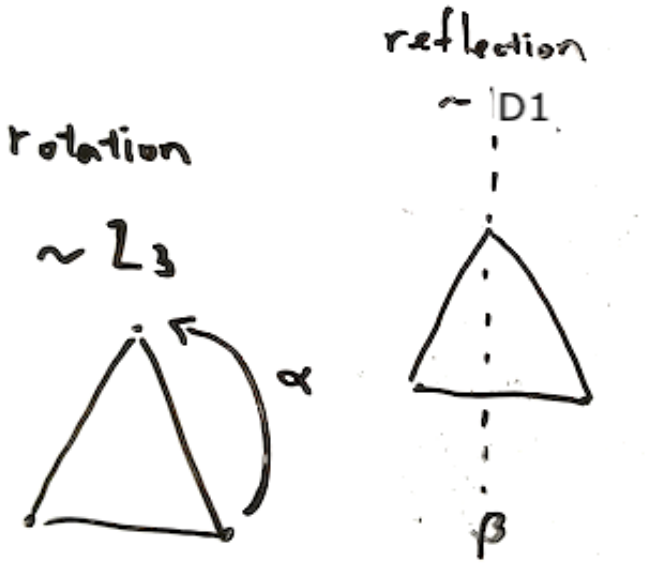
Product subgroup
(α^r)(β) = { 1, α^r, ..., β, $\alpha^r\beta$, ... } **self normal** if n odd (proof below)
(α^r)($\alpha\beta$) = { 1, α^r, ..., $\alpha\beta$, α^r+1β, ... } **normal** if r = 2
.... r variants

m = 1 (r = 0): reflection subgroup (see above)

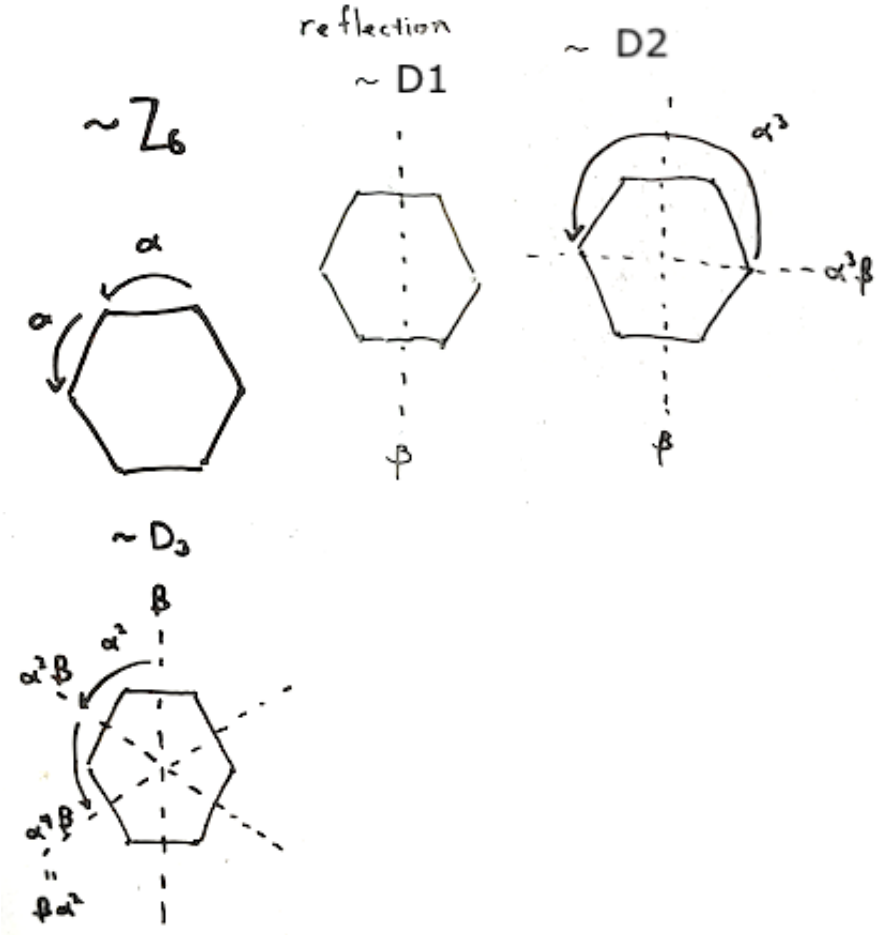
m = 2 (r = n/2, n even): D2
(α^n/2)(β)
(α^n/2)($\alpha\beta$)
.... r variants

m > 2: Dihedral subgroup Dm
(α^r)(β)
(α^r)($\alpha\beta$)
.... r variants

e.g. D3



e.g. D6



e.g. subgroups

D2	(α) = { 1, α }	~ Z2 = C2 normal
D3	(α)	~ Z3, normal
	(β)	~ D2 self-normal
	($\alpha\beta$)	~ D2 self-normal
	($\alpha^2\beta$)	~ D2 self-normal
D4	(α)	~ Z4 normal
	(α^2)	~ Z2 normal
	(β)	~ D2
	($\alpha\beta$)	~ D2
	...	
	(α^2) ∪ (β)	~ C2 x C2 normal
	(α^2) ∪ ($\alpha\beta$)	~ C2 x C2 normal
D5	(α)	~ Z5 normal
	(β)	~ D2 self-normal
	($\alpha\beta$)	~ D2 self-normal
	...	
D6	(α)	~ Z6 normal
	(α^2)	~ Z3 normal
	(α^3)	~ Z2 normal
	(β)	~ D2
	($\alpha\beta$)	~ D2

($\alpha^2\beta$) ~ D2

...

(α^3)(β) ~ C2 x C2

(α^3)($\alpha\beta$) ~ C2 x C2

...

(α^2)(β) ~ D3 normal

(α^2)($\alpha\beta$) ~ D3 normal

....

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proof of normality

If x = $\alpha^r\beta$

(α^k) x (α^-k) = α^2k x

($\alpha^k\beta$)x ($\alpha^k\beta$) = α^2(k-r) x

Hence

α^k $\in$ N(H) <=> **2k = 0 mod r**

i.e.

r even k = r/2, r, 3r/2, 2r, ...

r odd k = r, 2r, ...

OR

k = n/2

whereas

α^k, $\alpha^k\beta$ $\in$ H <=> k = 0 mod r

So

H is normal <=> α^k, $\alpha^k\beta$ $\in$ N(H) for all k

<=> r even with

k = r/2, r, 3r/2

= 1, 2, 3, ...

<=> r = 2

H is self-normal <=> α^k $\in$ N(H) only if k = 0 mod r

<=> r = n/m odd and n/2 ≠ 0 mod r

<=> if n even n/m and m odd

n odd n/m odd

<=> n odd

QED

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Normal subgroups

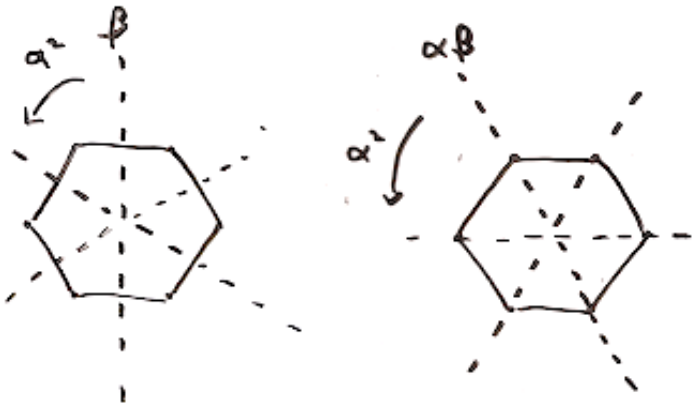
There are

rotation group (α^m) for m | n

dihedral subgroups ~ D(n/2) if n is even

{ 1, α^2, ..., β, $\alpha^2\beta$, ... } reflection axis through vertices

= 1^Dn ∪ (α^2)^Dn ∪ β^Dn
{ 1, α^2, ..., $\alpha\beta$, $\alpha^3\beta$, ... } reflection axis through sides
= 1^Dn ∪ (α^2)^Dn ∪ $\alpha\beta$^Dn



Hence

of dihedral subgroups =

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proof of normality

claim: H ~ D(n/m) subgroup is normal only if m = 2
Consider conjugation

$\alpha\ \alpha^m\beta\ \alpha$^-1 = α^m ($\alpha\beta\ \alpha$^-1)
= α^m $\alpha^2\beta$
= α^(m+2) β
=> α^(m+2) $\beta \in$ H
=> only m = 2 is possible

QED
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This
is
code
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